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(54) **TRANSISTOR SHUTDOWN CIRCUT**

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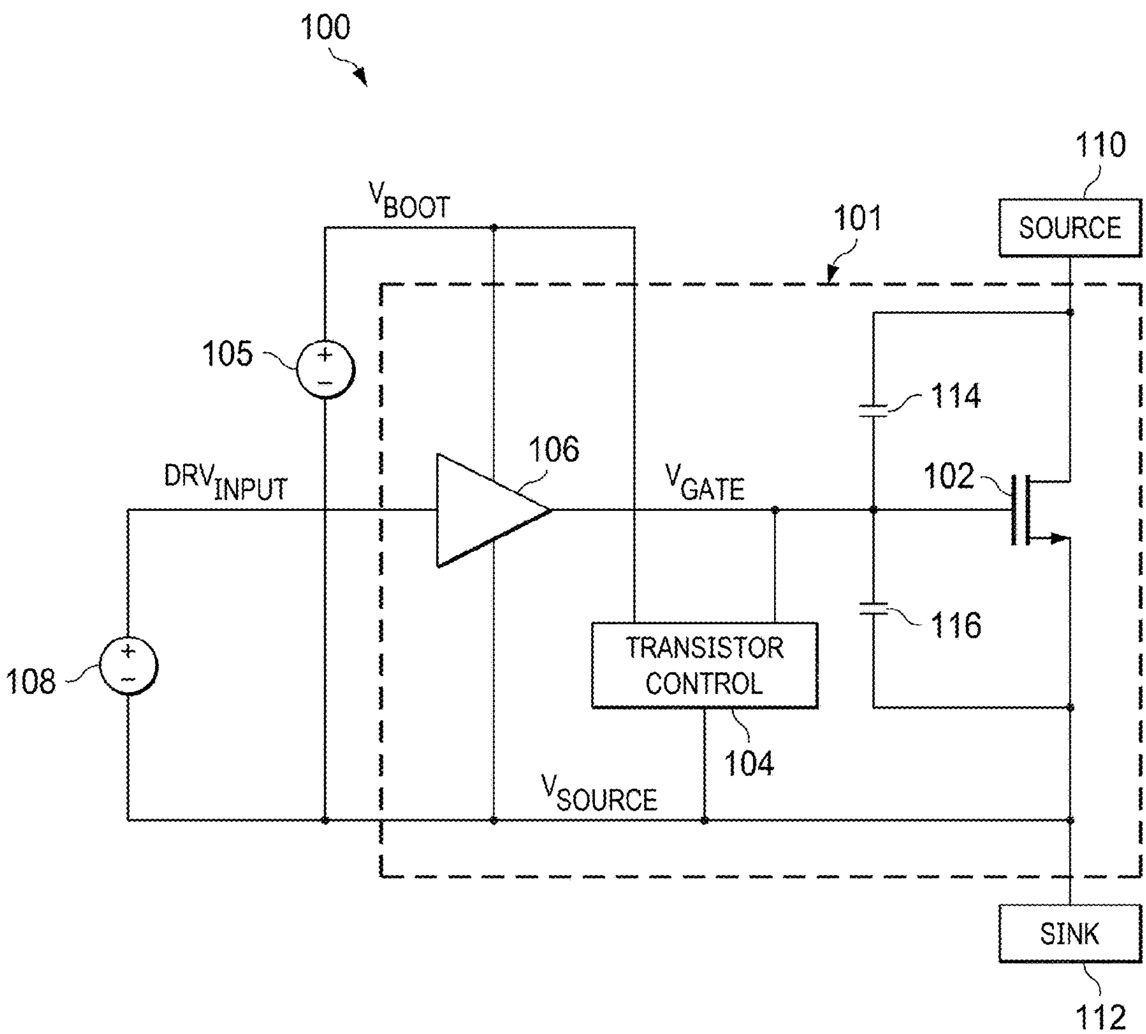
(57) **ABSTRACT**

A circuit includes a first transistor, a second transistor, a third transistor, and a fourth transistor. The first transistor has a first terminal, a second terminal, and a control terminal. The second transistor has a first terminal coupled to the control terminal of the first transistor, a second terminal coupled to the second terminal of the first transistor, and a control terminal. The third transistor has a first terminal coupled to the control terminal of the first transistor, a second terminal coupled to the control terminal of the second transistor, and a control terminal. The fourth transistor has a first terminal configured to receive a drive voltage, a second terminal coupled to the control terminal of the third transistor, and a control terminal coupled to the first terminal of the fourth transistor.



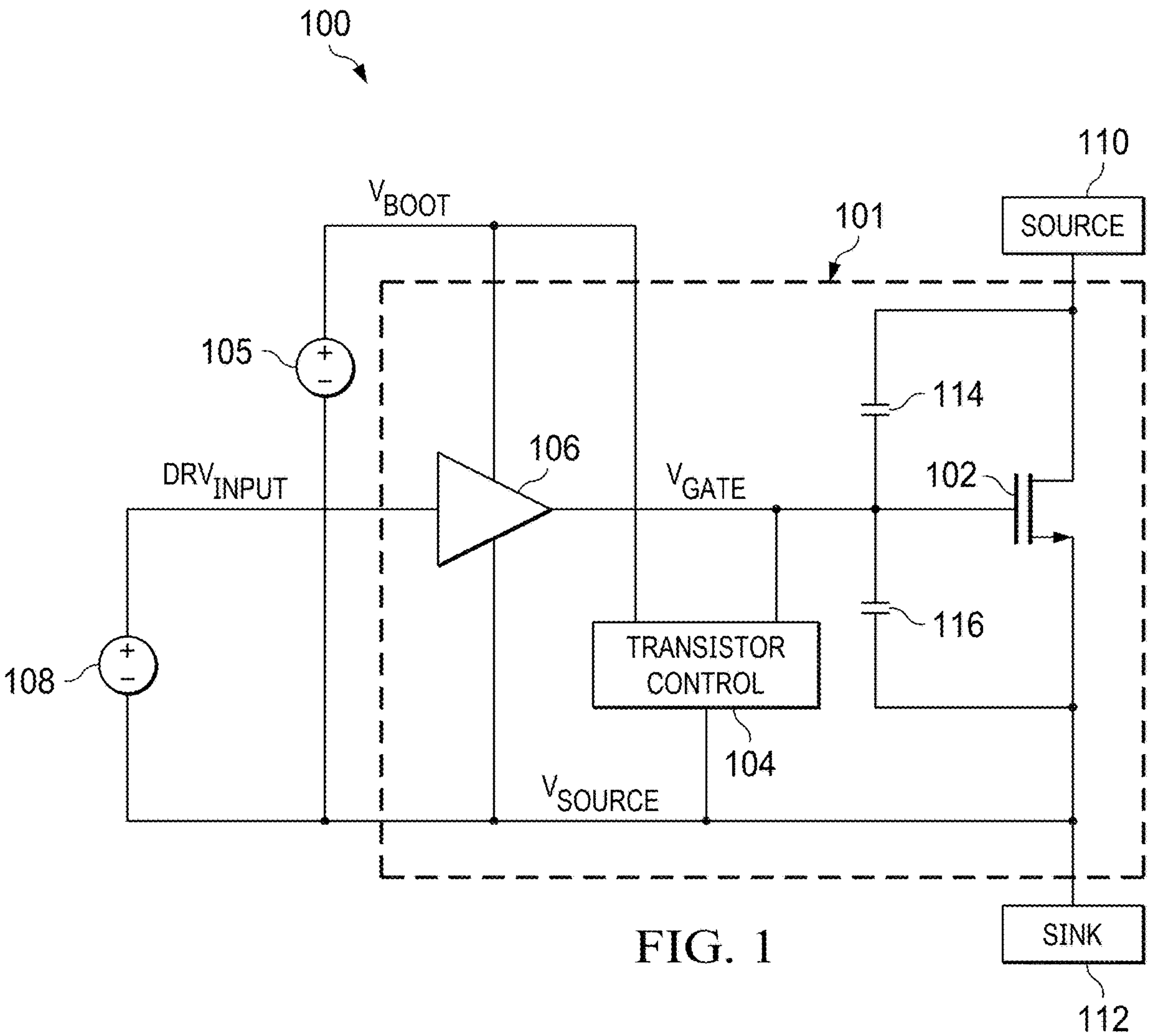
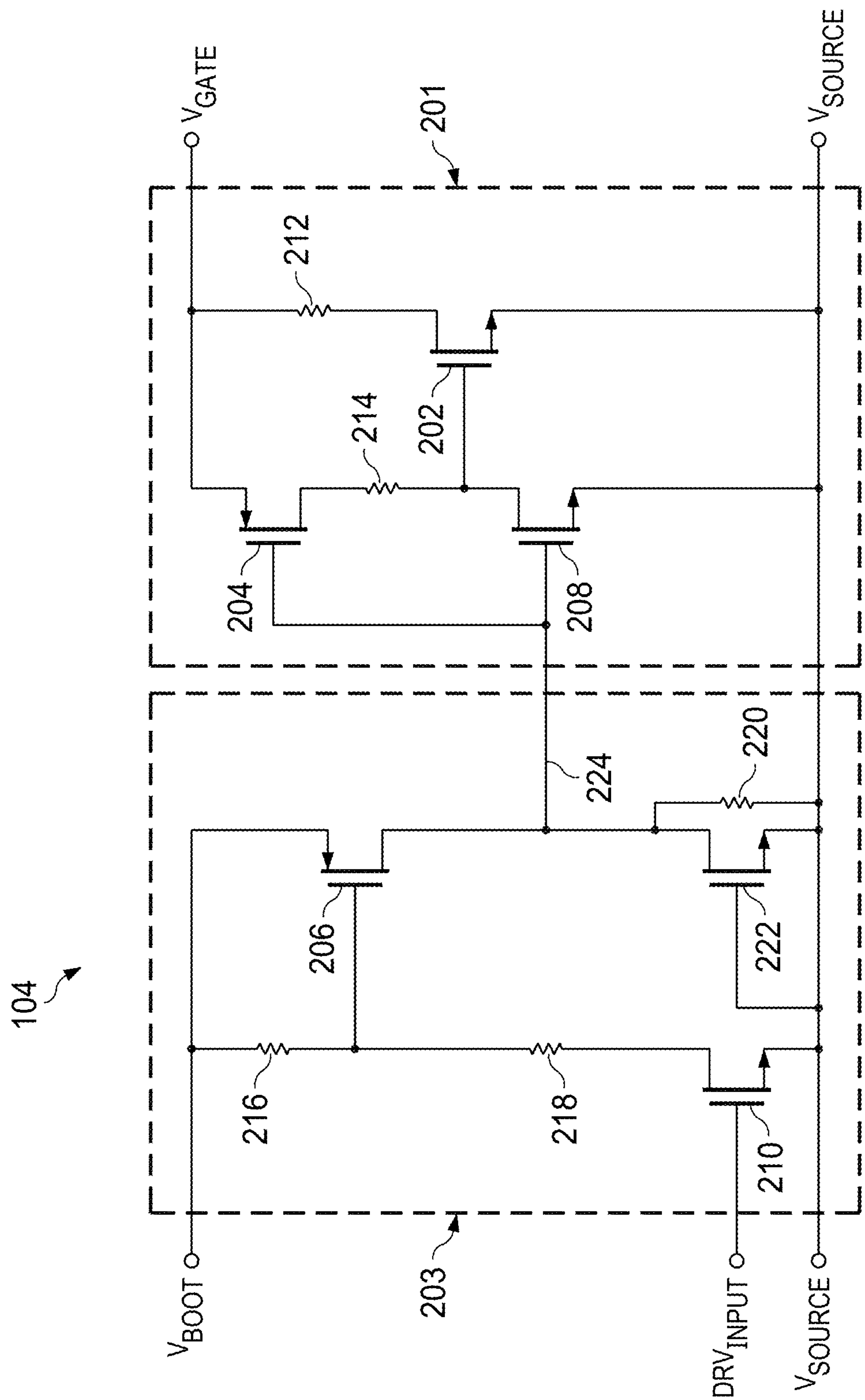

100
110
SOURCE
V_BOOT
101
105
114
106
DRV_INPUT
V_GATE
102
116
108
TRANSISTOR CONTROL
104
V_SOURCE
SINK
112

FIG. 1



V_{GATE}
V_{SOURCE}
201
212
214
202
204
208
224
220
222
206
104
216
218
210
203
V_{BOOT}
DRV_{INPUT}
V_{SOURCE}

FIG. 2

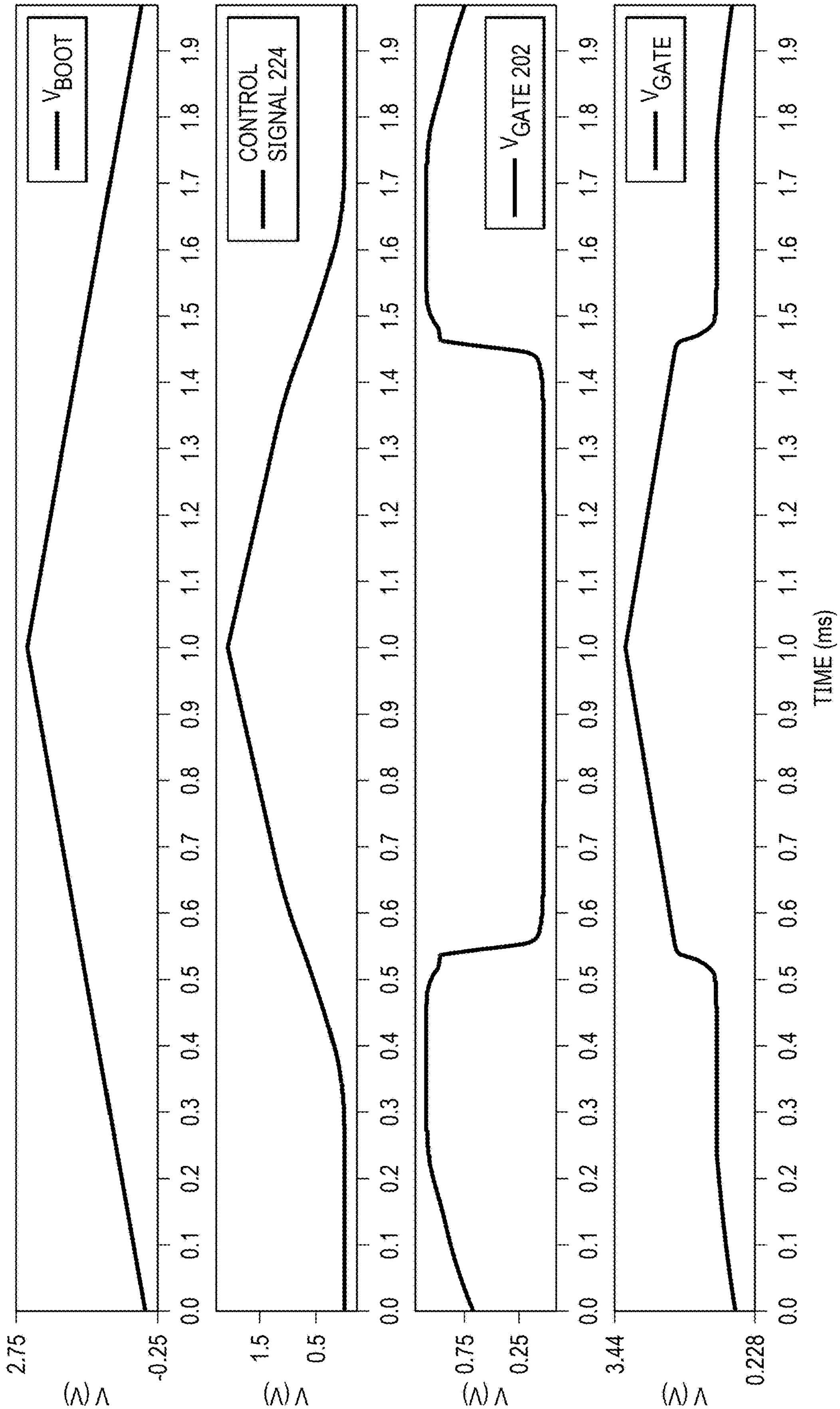

V (V)
2.75
-0.25
V_{BOOT}
1.5
0.5
CONTROL SIGNAL 224
0.75
0.25
V_{GATE} 202
3.44
0.228
V_{GATE}
0.0 0.1 0.2 0.3 0.4 0.5 0.6 0.7 0.8 0.9 1.0 1.1 1.2 1.3 1.4 1.5 1.6 1.7 1.8 1.9
TIME (ms)

FIG. 3

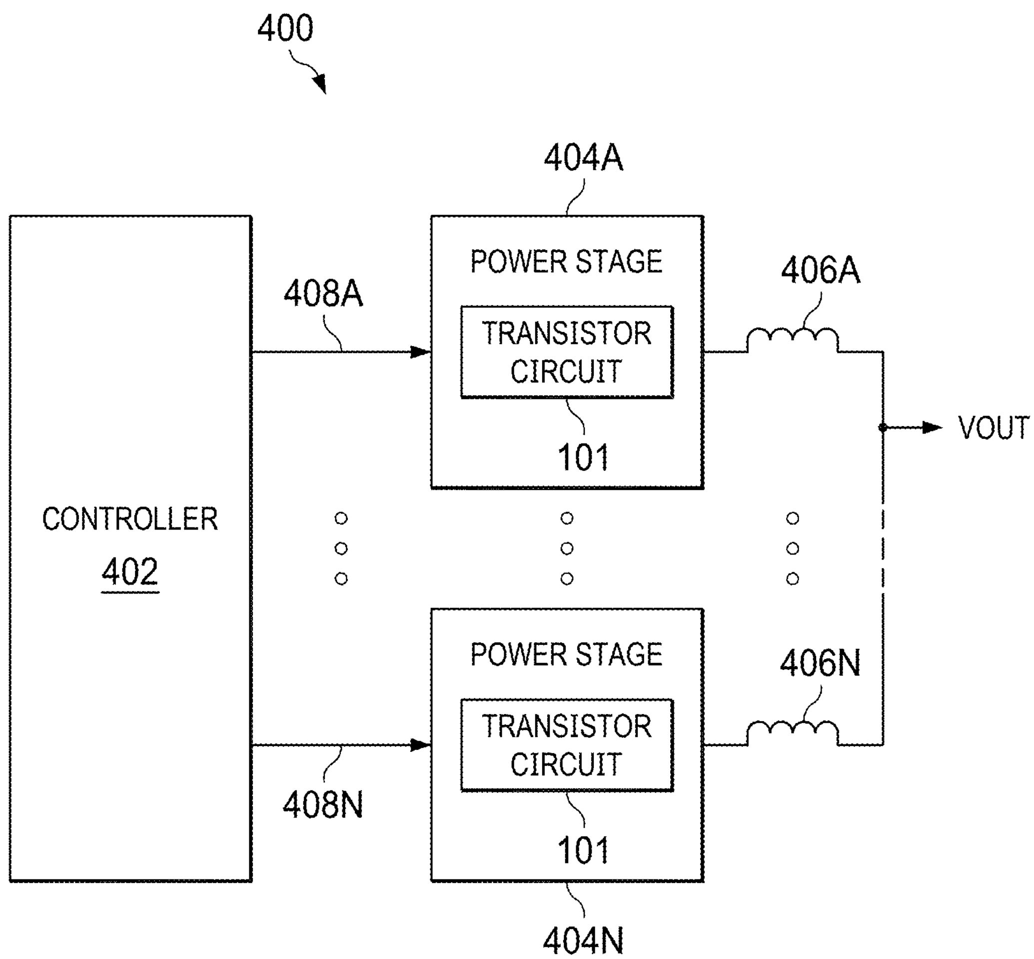

400
404A
POWER STAGE
TRANSISTOR CIRCUIT
101
406A
408A
VOUT
CONTROLLER
402
POWER STAGE
TRANSISTOR CIRCUIT
101
406N
408N
404N

FIG. 4

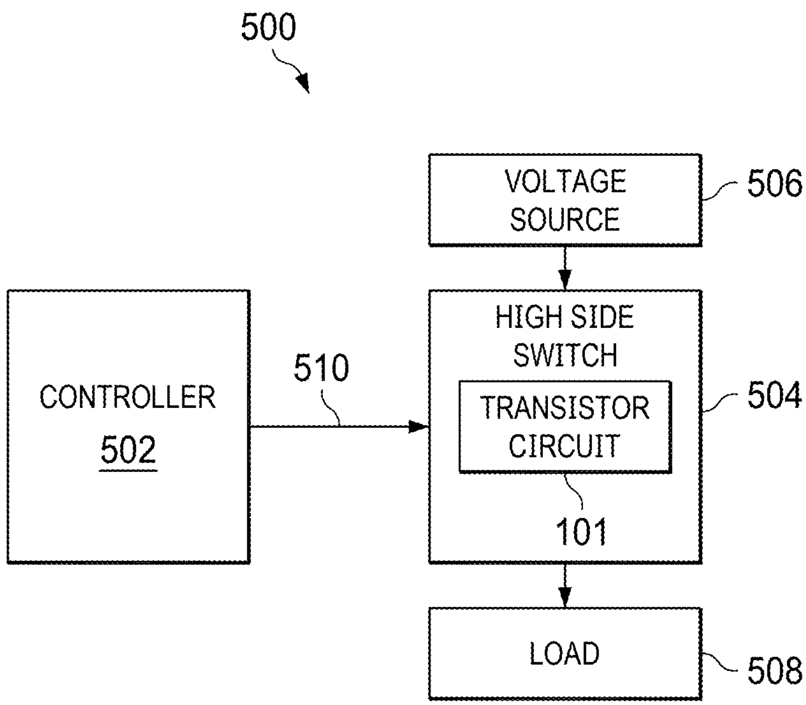

500
VOLTAGE SOURCE
506
HIGH SIDE SWITCH
TRANSISTOR CIRCUIT
101
504
CONTROLLER
502
510
LOAD
508

FIG. 5

TRANSISTOR SHUTDOWN CIRCUT

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application claims priority to U.S. Provisional Application No. 63/562,518, filed Mar. 7, 2024, entitled "Low IQ and Low Area Shut Down Scheme for Deterministic FET Power-Off in Switching Regulator Applications", which is hereby incorporated herein by reference in its entirety.

BACKGROUND

[0002] A switch can be used to control the flow of current from a voltage source to a load circuit. A transistor can be used to implement such as switch. A circuit using a transistor as a switch may include a gate driver to control the transistor. The gate driver provides a control signal tor turning the transistor on and off.

SUMMARY

[0003] In one example, a circuit includes a drive control circuit and a threshold circuit. The drive control circuit has a first terminal configured to provide a transistor control signal, and a second terminal configured to receive a transistor reference signal. The drive control circuit includes a first transistor and a second transistor. The first transistor has a first terminal coupled to the first terminal of the drive control circuit, a second terminal coupled to the second terminal of the drive control circuit, and a control terminal. The second transistor has a first terminal coupled to the first terminal of the first transistor, a second terminal coupled to the control terminal of the first transistor, and a control terminal. The threshold circuit has a first terminal configured to receive a drive voltage. The threshold circuit includes a third transistor having a first terminal coupled to the first terminal of the threshold circuit, a second terminal coupled to the control terminal of the second transistor, and a control terminal coupled to the first terminal of the third transistor.

[0004] In another example, a circuit includes a first transistor, a second transistor, a third transistor, and a fourth transistor. The first transistor has a first terminal, a second terminal, and a control terminal. The second transistor has a first terminal coupled to the control terminal of the first transistor, a second terminal coupled to the second terminal of the first transistor, and a control terminal. The third transistor has a first terminal coupled to the control terminal of the first transistor, a second terminal coupled to the control terminal of the second transistor, and a control terminal. The fourth transistor has a first terminal configured to receive a drive voltage, a second terminal coupled to the control terminal of the third transistor, and a control terminal coupled to the first terminal of the fourth transistor.

[0005] In a further example, a circuit includes a transistor, a driver circuit, and a transistor control circuit. The transistor has a first terminal, a second terminal, and a control terminal. The driver circuit has an output coupled to the control terminal of the transistor, an input configured to receive a drive signal, and a terminal configured to receive a drive voltage. The transistor control circuit has a first terminal coupled to the control terminal of the transistor, a second terminal coupled to the second terminal of the transistor, and a third terminal coupled to the terminal of the driver circuit. The transistor control circuit is configured to conduct current between the control terminal of the transistor and the second terminal of the transistor responsive to the drive voltage being less than a threshold.

BRIEF DESCRIPTION OF THE DRAWINGS

[0006] FIG. **1** is a block diagram of an example transistor circuit that includes a transistor control circuit.

[0007] FIG. **2** is a schematic diagram of an example transistor control circuit suitable for use in the transistor circuit of FIG. **1**.

[0008] FIG. **3** is a graph of example signals in the transistor control circuit of FIG. **2**.

[0009] FIG. **4** is a block diagram of an example multiphase switching converter that includes the transistor control circuit of FIG. **2**.

[0010] FIG. **5** is a block diagram of an example circuit that includes a high-side switch using the transistor control circuit of FIG. **2**.

DETAILED DESCRIPTION

[0011] A transistor (e.g., a metal oxide semiconductor field effect transistor (MOSFET)) can be used as a switch in some applications, with gate driver circuitry to turn the transistor on and off. If the power supply voltage provided to the gate driver circuitry is insufficient to properly power the gate driver circuitry, then the transistor may be improperly controlled. For example, the gate of a MOSFET may be left floating, and the transistor may be inadvertently turned on by transients caused by circuits coupled to the transistor. Such uncontrolled turn-on of the transistor can result in the flow of currents that may damage the transistor or circuits coupled to the transistor. The transistor control circuits described herein prevent unintended turn on the transistor if the driver circuitry is not properly powered. Accordingly, the transistor control circuits can prevent damage to the transistor or other circuits caused by uncontrolled current flow.

[0012] FIG. 1 is a block diagram of an example circuit **100** that includes transistor shutdown based on gate driver supply voltage. The circuit **100** includes a transistor circuit **101**, a voltage source circuit **110**, a current sink circuit **112**, a boot voltage source **105**, and a drive signal source **108**. The transistor circuit **101** includes a transistor **102**, a transistor control circuit **104**, and a driver circuit **106**. The transistor **102** conducts current from the voltage source circuit **110** to the current sink circuit **112** if the transistor **102** is turned on. The transistor **102** may be an n-channel field effect transistor (NFET) is some examples of the transistor circuit **101**. The transistor **102** has a first terminal (e.g., drain) coupled to an output of the voltage source circuit **110**, a second terminal (e.g., source) coupled to an input of the current sink circuit **112**, and a control terminal (e.g., gate). Gate-to-drain capacitance of the transistor **102** is shown as capacitor **114**, and gate-to-source capacitance of the transistor **102** is shown as capacitor **116**. The voltage source circuit **110** may be a battery, a battery pack, a power supply, or other circuit that provides an output voltage. The current sink circuit **112** may be power supply circuitry, an application circuit, or other circuit that sinks current from the voltage source circuit **110** through the transistor **102**.

[0013] The driver circuit **106** controls the transistor **102**. The driver circuit **106** has an input coupled to an output of the drive signal source **108**, and an output coupled to the control terminal of the transistor **102**. The driver circuit **106**

receives a driver input signal (DRV_{INPUT}) from the drive signal source **108**, and provides the drive signal (V_{GATE}) to the control terminal of the transistor **102** with voltage and current suitable for proper operation of the transistor **102**. A power terminal of the driver circuit **106** is coupled to an output of the boot voltage source **105** for receipt of voltage (V_{BOOT}) that powers the driver circuit **106**. A reference terminal of the driver circuit **106** is coupled to the second terminal of the transistor **102** for receipt of reference voltage (V_{SOURCE}). The boot voltage source **105** may be a charge pump, a fly-back converter, or other circuit that generates voltage for driving the transistor **102**. The drive signal source **108** may be a pulse width modulation circuit or other circuit that generates a signal for controlling the transistor **102**.

[0014] If the voltage powering the driver circuit **106** (V_{BOOT}) is insufficient to properly power the circuitry of the driver circuit **106**, then the output of the driver circuit **106** may have high-impedance, and the control terminal of the transistor **102** is left floating. With the control terminal of the transistor **102** floating, the transistor **102** can be turned on through the capacitor **114** or the capacitor **116** by transients caused by the voltage source circuit **110** or the current sink circuit **112**. Inadvertent turn on the transistor **102** can result in large current flow through the transistor **102** that can damage the transistor **102**, the voltage source circuit **110**, or the current sink circuit **112**.

[0015] The transistor control circuit **104** prevents unintended turn on the transistor **102** if the driver circuit **106** is not properly powered. The transistor control circuit **104** has a terminal coupled to the power terminal of the driver circuit **106**, a terminal coupled to the reference terminal of the driver circuit **106**, and a terminal coupled to the control terminal of the transistor **102**. The transistor control circuit **104** conducts current from the control terminal of the transistor **102** to the second terminal of the transistor **102** to turn off the transistor **102** if V_{BOOT} is too low to properly power the circuitry of the driver circuit **106** (e.g., less than 1 volt). If V_{BOOT} is high enough to properly power the circuitry of the driver circuit **106** (e.g., greater than 1 volt), then the transistor control circuit **104** releases the control terminal of the transistor **102** to be controlled by the driver circuit **106**.

[0016] FIG. **2** is a schematic diagram of an example transistor control circuit **104**. The transistor control circuit **104** includes a drive control circuit **201** and a threshold circuit **203**. The threshold circuit **203** has a first terminal coupled to the power terminal of the driver circuit **106** for receipt of V_{BOOT}, a second terminal coupled to the reference terminal of the driver circuit **106** for receipt of V_{SOURCE}, an input coupled to the input of the driver circuit **106** for receipt of DRV_{INPUT}, and an output for providing a control signal **224** to the drive control circuit **201**. The threshold circuit **203** provides the control signal **224** with a first voltage (e.g., less than the threshold voltage of the transistor **208**) to activate the drive control circuit **201**. Activating the drive control circuit **201** causes the drive control circuit **201** to conduct current from the control terminal of the transistor **102** to the second terminal of the transistor **102**. The drive control circuit **201** provides the control signal **224** with a second voltage (e.g., greater than the threshold voltage of the transistor **208**) to deactivate the drive control circuit **201**. Deactivating the drive control circuit **201** causes the drive control circuit **201** to disconnect the control terminal of the transistor **102** from the second terminal of the transistor **102**, which allows the driver circuit **106** to control the transistor **102**.

[0017] The threshold circuit **203** includes a transistor **206**, a transistor **210**, a resistor **216**, a resistor **218**, and a resistor **220**. The threshold circuit **203** may also include a transistor **222**. The transistor **210** and the transistor **222** may be NFETs. The transistor **206** may be a p-channel field effect transistor (PFET). If V_{BOOT} is less than a threshold (e.g., less than 1 volt), then the threshold circuit **203** provides the control signal **224** with the first voltage, to activate the drive control circuit **201**. If V_{BOOT} is greater than the threshold (e.g., greater than 1 volt), then the threshold circuit **203** provides the control signal **224** with the second voltage, to deactivate the drive control circuit **201**.

[0018] The transistor **206** has a first terminal (e.g., source) coupled to the first terminal of the threshold circuit **203**, a second terminal coupled to the drive control circuit **201**, and a control terminal coupled to the first terminal of the transistor **206** via the resistor **216**. A first terminal of the resistor **216** is coupled to the control terminal of the transistor **206** and a second terminal of the resistor **216** is coupled to the first terminal of the transistor **206**. The resistor **220** has a first terminal coupled to the second terminal of the transistor **206**, and a second terminal coupled to the second terminal of the threshold circuit **203**. The transistor **210** has a first terminal (e.g., drain) coupled to the control terminal of the transistor **206** via the resistor **218**, a second terminal coupled to the second terminal of the threshold circuit **203**, and a control terminal coupled to the input of the threshold circuit **203** and the input of the driver circuit **106**.

[0019] If the voltage of DRV_{INPUT} received at the control terminal of the transistor **210** is too low to turn on the transistor **210** (e.g., DRV_{INPUT} has logic low state), then the transistor **206** is turned off, and the resistor **220** pulls the control signal **224** output by the threshold circuit **203** down to activate the drive control circuit **201**. The value of V_{BOOT} needed to turn on the transistor **206** is determined by the resistors **216**, **218**, and **220**, and the transistor **206**, various combinations of which may be selected to provide a desired threshold voltage. If the voltage of DRV_{INPUT} is sufficient to turn on the transistor **210**, then current flows through the resistor **216** and the resistor **218**, and the voltage at the control terminal of the transistor **206** falls. If the voltage at the control terminal of the transistor **206** causes the transistor **206** to turn on, then current flows through the transistor **206** and the resistor **220**, and a voltage is developed across the resistor **220** to provide the control signal **224** with the second voltage.

[0020] The transistor **222** may be coupled in parallel with the resistor **220** between the second terminal of the transistor **206** and the second terminal of the threshold circuit **203**. The transistor **222** has a first terminal (e.g., drain coupled to the second terminal of the transistor **206**, a second terminal (e.g., source) coupled to the second terminal of the threshold circuit **203**, and a control terminal (e.g., gate) coupled to the second terminal of the threshold circuit **203**. The transistor **222** is biased off, and the body diode of the transistor **222** is coupled between the second terminal of the transistor **206** and the second terminal of the threshold circuit **203** for electrostatic discharge protection.

[0021] The drive control circuit **201** has a first terminal coupled to the control terminal of the transistor **102**, and a second terminal coupled to the second terminal of the transistor **102**. The drive control circuit **201** includes a transistor **202**, a transistor **204**, a transistor **208**, a resistor **212**, and a resistor **214**. The transistor **202** and the transistor **208** may be NFETs, and the transistor **204** may be a PFET. The transistor **202** has a first terminal (e.g., drain) coupled to the first terminal of the drive control circuit **201** (and the control terminal of the transistor **102**) via the resistor **212**, a second terminal (e.g., source) coupled to the second terminal of the drive control circuit **201** (and the second terminal of the transistor **102**), and control terminal (e.g., gate). A first terminal of the resistor **212** is coupled to the first terminal of the transistor **202** (for providing V_{GATE}), and a second terminal of the resistor **212** is coupled to the first terminal of the drive control circuit **201**. The transistor **204** has a first terminal (e.g., source) coupled to the second terminal of the resistor **212**, a second terminal (e.g., drain) coupled to the control terminal of the resistor **212** via the resistor **214**, and control terminal (e.g., gate) coupled to the output of the threshold circuit **203**. The resistor **214** has a first terminal coupled to the second terminal of the transistor **204**, and a second terminal coupled to the control terminal of the transistor **202**. If the transistor **204** is turned on, then the transistor **202** is diode-connected between the control terminal of the transistor **102** and the second terminal of the transistor **102**, and can conduct current from the control terminal of the transistor **102** to the second terminal of the transistor **102**. Thus, the control terminal of the transistor **102** is pulled-down by the transistor control circuit **104** if the control signal **224** has the first voltage.

[0022] The transistor **208** has a first terminal (e.g., drain) coupled to the control terminal of the transistor **202**, a second terminal (e.g., source) coupled to the second terminal of the transistor **202**, and a control terminal (e.g., gate) coupled to the control terminal of the transistor **204**. If the control signal **224** provided by the threshold circuit **203** has the first voltage, then the transistor **204** is turned on and the transistor **208** is turned off to diode-connect the transistor **202**, and turn off the transistor **102**. If the control signal **224** provided by the threshold circuit **203** has the second voltage, then transistor **208** is turned on, and the transistors **208** and **202** are turned off to allow the driver circuit **106** to control the transistor **102**.

[0023] FIG. **3** is a graph of example signals in the transistor circuit **101**. FIG. **3** shows the drive voltage V_{BOOT} that powers the driver circuit **106**, the control signal **224** provided by the threshold circuit **203**, voltage at the control terminal of the transistor **202** ($V_{GATE202}$), and voltage at the control terminal of the transistor **102** (V_{GATE}). V_{BOOT} is shown ramping from about zero volts to about 2.75 volts and back to about zero volts to illustrate operation of the transistor control circuit **104** with changing V_{BOOT}. At time zero of the graph, the transistor **202** is off, the transistor **208** is off, the transistor **204** is on, and the transistor **206** is on. Accordingly, the transistor **202** is diode-connected through the transistor **204** and the resistor **214**, and the transistor **102** is turned off. As the voltage of V_{BOOT} increases, the transistor **206** turns on, and the voltage across the resistor **220** (voltage of the control signal **224**) increases. When the control signal **224** exceeds the threshold voltage of the transistor **208**, the transistor **208** turns on, the transistor **204** turns off, and $V_{GATE202}$ falls to turn off the transistor **202**. With the transistor **202** turned off the driver circuit **106** controls the transistor **102**, and V_{GATE} rises and follows V_{BOOT} to turn on the transistor **102**.

[0024] As V_{BOOT} falls from about 2.75 volts back to about zero volts, the control signal **224** falls. When the control signal **224** is less than the threshold voltage of the transistor **208**, the transistor **208** turns off the transistor **204** turns on, and $V_{GATE202}$ rises to turn on the transistor **202**. With the transistor **202** turned on the transistor control circuit **104** controls the transistor **102**, and V_{GATE} falls to turn off the transistor **102**.

[0025] FIG. **4** is a block diagram of an example multi-phase switching converter **400** that includes transistor shutdown based on gate driver supply voltage. The multi-phase switching converter **400** includes a controller **402**, power stages **404**A through **404**N, and inductors **406**A through **406**N. "N" denotes the total number of power stages included in the multi-phase switching converter **400**. Each of the power stages **404**A through **404**N is coupled to an output of the controller **402**. The controller **402** is an example of the drive signal source **108**, and generates the drive signals **408**A through **408**N. A first terminal of each inductor **406**A through **4066**N is coupled to an output of one of the power stages **404**A through **404**N. The second terminals of the inductors **406**A through **406**N are connected to provide the output voltage VOUT for powering a load circuit.

[0026] Each of the power stages **404**A through **404**N includes an example of the transistor circuit **101**. In an implementation of the multi-phase switching converter **400**, the transistor circuit **101** may be provided as a high-side switching circuit of a buck converter. Each power stage **404**A through **404**N may include low-side switching circuitry that has been omitted from FIG. **4** in the interest of clarity. The transistor circuit **101** includes the transistor **102**, the transistor control circuit **104** and the driver circuit **106**. Considering the power stage **404**A, the input of the driver circuit **106** receives the drive signal **408**A, and drives the drive signal **408**A to the transistor **102**.

[0027] As explained with regard to FIG. **2**, the transistor control circuit **104** controls the transistor **102** if the drive voltage powering the driver circuit **106** is below a threshold (e.g., 1 volt). The transistor control circuit **104** prevents the transistor **102** from turning on inadvertently if the driver circuit **106** is unable to control the transistor **102**. By preventing inadvertent turn-on of the transistor **102**, the transistor control circuit **104** prevents the flow of large currents through transistor **102** that may damage the transistor **102** or components coupled to the transistor **102**. When the drive voltage is high enough to properly power the driver circuit **106** (e.g., greater than 1 volt), the transistor control circuit **104** relinquishes control of the transistor **102** to the driver circuit **106**.

[0028] Each of the power stages **404**A through **404**N may also include a low-side transistor (e.g., an NFET) coupled to the transistor circuit **101** as the current sink circuit **112** (shown in FIG. **1**). For example, a drain of the low-side transistor may be coupled to V_{SOURCE}, a source of the low-side transistor may be coupled to ground, and a gate of the low-side transistor may be coupled to an output of a low-side transistor driver circuit (not shown). The low-side transistors have been omitted from the power stages **404**A through **404**N in the interest of clarity.

[0029] FIG. **5** is a block diagram of an example circuit **500** that includes a high-side switching. The circuit **500** includes a controller **502**, a high-side switch circuit **504**, a voltage source **506**, and a load circuit **508**. The high-side switch circuit **504** is coupled to the controller **502**, the voltage source **506**, and the load circuit **508**. The controller **502** provides a control signal **510** for controlling the high-side switch circuit **504**. The controller **502** is an example of the drive signal source **108**. The high-side switch circuit **504** conducts current from the voltage source **506** to the load circuit **508**. The voltage source **506** may be a battery, a power supply, or other device. The load circuit **508** may be any circuit that is powered by the voltage source **506**.

[0030] The high-side switch circuit **504** includes an example of the transistor circuit **101**. As explained with regard to FIG. **2**, the transistor circuit **101** includes the transistor **102**, the transistor control circuit **104** and the driver circuit **106**. The transistor **102** conducts current from the voltage source **506** to the load circuit **508**. The driver circuit **106** drives the control signal **510** to the transistor **102**. The transistor control circuit **104** controls the transistor **102** if the drive voltage powering the driver circuit **106** is below a threshold (e.g., 1 volt). The transistor control circuit **104** prevents the transistor **102** from turning on inadvertently if the driver circuit **106** is unable to control the transistor **102**. By preventing inadvertent turn-on of the transistor **102**, the transistor control circuit **104** prevents the flow of large currents through transistor **102** that may damage the transistor **102** or the load circuit **508**. When the drive voltage is high enough to properly power the driver circuit **106** (e.g., greater than 1 volt), the transistor control circuit **104** relinquishes control of the transistor **102** to the driver circuit **106**.

[0031] In this description, the term "couple" may cover connections, communications, or signal paths that enable a functional relationship consistent with this description. For example, if device A generates a signal to control device B to perform an action: (a) in a first example, device A is coupled to device B by direct connection; or (b) in a second example, device A is coupled to device B through intervening component C if intervening component C does not alter the functional relationship between device A and device B, such that device B is controlled by device A via the control signal generated by device A.

[0032] Also, in this description, the recitation "based on" means "based at least in part on." Therefore, if X is based on Y, then X may be a function of Y and any number of other factors.

[0033] A device that is "configured to" perform a task or function may be configured (e.g., programmed and/or hardwired) at a time of manufacturing by a manufacturer to perform the function and/or may be configurable (or reconfigurable) by a user after manufacturing to perform the function and/or other additional or alternative functions. The configuring may be through firmware and/or software programming of the device, through a construction and/or layout of hardware components and interconnections of the device, or a combination thereof.

[0034] As used herein, the terms "terminal," "node," "interconnection," "pin" and "lead" are used interchangeably. Unless specifically stated to the contrary, these terms are generally used to mean an interconnection between or a terminus of a device element, a circuit element, an integrated circuit, a device or other electronics or semiconductor component.

[0035] A circuit or device that is described herein as including certain components may instead be adapted to be coupled to those components to form the described circuitry or device. For example, a structure described as including one or more semiconductor elements (such as transistors), one or more passive elements (such as resistors, capacitors, and/or inductors), and/or one or more sources (such as voltage and/or current sources) may instead include only the semiconductor elements within a single physical device (e.g., a semiconductor die and/or integrated circuit (IC) package) and may be adapted to be coupled to at least some of the passive elements and/or the sources to form the described structure either at a time of manufacture or after a time of manufacture, for example, by an end-user and/or a third-party.

[0036] While the use of particular transistors is described herein, other transistors (or equivalent devices) may be used instead with little or no change to the remaining circuitry. For example, a field effect transistor ("FET") (such as an n-channel FET (NFET) or a p-channel FET (PFET)), a bipolar junction transistor (BJT—e.g., NPN transistor or PNP transistor), an insulated gate bipolar transistor (IGBT), and/or a junction field effect transistor (JFET) may be used in place of or in conjunction with the devices described herein. The transistors may be depletion mode devices, drain-extended devices, enhancement mode devices, natural transistors, or other types of device structure transistors. Furthermore, the devices may be implemented in/over a silicon substrate (Si), a silicon carbide substrate (SiC), a gallium nitride substrate (GaN) or a gallium arsenide substrate (GaAs).

[0037] References may be made in the claims to a transistor's control input and its current terminals. In the context of a FET, the control input is the gate, and the current terminals are the drain and source. In the context of a BJT, the control input is the base, and the current terminals are the collector and emitter.

[0038] References herein to a FET being "ON" or "enabled" means that the conduction channel of the FET is present and drain current may flow through the FET. References herein to a FET being "OFF" or "disabled" means that the conduction channel is not present so drain current does not flow through the FET. An "OFF" FET, however, may have current flowing through the transistor's body-diode.

[0039] Circuits described herein are reconfigurable to include additional or different components to provide functionality at least partially similar to functionality available prior to the component replacement. Components shown as resistors, unless otherwise stated, are generally representative of any one or more elements coupled in series and/or parallel to provide an amount of impedance represented by the resistor shown. For example, a resistor or capacitor shown and described herein as a single component may instead be multiple resistors or capacitors, respectively, coupled in parallel between the same nodes. For example, a resistor or capacitor shown and described herein as a single component may instead be multiple resistors or capacitors, respectively, coupled in series between the same two nodes as the single resistor or capacitor.

[0040] While certain elements of the described examples are included in an integrated circuit and other elements are external to the integrated circuit, in other example embodiments, additional or fewer features may be incorporated into the integrated circuit. In addition, some or all of the features illustrated as being external to the integrated circuit may be included in the integrated circuit and/or some features illustrated as being internal to the integrated circuit may be incorporated outside of the integrated. As used herein, the term "integrated circuit" means one or more circuits that are: (i) incorporated in/over a semiconductor substrate; (ii) incorporated in a single semiconductor package; (iii) incorporated into the same module; and/or (iv) incorporated in/on the same printed circuit board.

[0041] Uses of the phrase "ground" in the foregoing description include a chassis ground, an Earth ground, a floating ground, a virtual ground, a digital ground, a common ground, and/or any other form of ground connection applicable to, or suitable for, the teachings of this description. In this description, unless otherwise stated, "about," "approximately" or "substantially" preceding a parameter means being within +/−10 percent of that parameter or, if the parameter is zero, a reasonable range of values around zero.

[0042] Modifications are possible in the described embodiments, and other embodiments are possible, within the scope of the claims.

What is claimed is:

1. A circuit comprising:

a drive control circuit having a first terminal configured to provide a transistor control signal, and a second terminal configured to receive a transistor reference signal, the drive control circuit including:

a first transistor having a first terminal coupled to the first terminal of the drive control circuit, a second terminal coupled to the second terminal of the drive control circuit, and a control terminal; and

a second transistor having a first terminal coupled to the first terminal of the first transistor, a second terminal coupled to the control terminal of the first transistor, and a control terminal; and

a threshold circuit having a first terminal configured to receive a drive voltage, the threshold circuit including a third transistor having a first terminal coupled to the first terminal of the threshold circuit, a second terminal coupled to the control terminal of the second transistor, and a control terminal coupled to the first terminal of the third transistor.

2. The circuit of claim **1**, wherein the drive control circuit includes a fourth transistor having a first terminal coupled to the control terminal of the first transistor, a second terminal coupled to the second terminal of the first transistor, and control terminal coupled to the control terminal of the second transistor.

3. The circuit of claim **1**, wherein the drive control circuit includes a resistor having a first terminal coupled to the first terminal of the first transistor and a second terminal coupled to the first terminal of the second transistor.

4. The circuit of claim **1**, wherein the drive control circuit includes a resistor having a first terminal coupled to the second terminal of the second transistor and a second terminal coupled to the control terminal of the first transistor.

5. The circuit of claim **1**, wherein:

the threshold circuit has a second terminal configured to receive a driver input signal; and

the threshold circuit includes a fourth transistor having a first terminal coupled to the control terminal of the third transistor, a second terminal coupled to the second terminal of the first transistor, and a control terminal coupled to the second terminal of the threshold circuit.

6. The circuit of claim **5**, wherein the threshold circuit includes a resistor having a first terminal coupled to the control terminal of the third transistor, and a second terminal coupled to the first terminal of the fourth transistor.

7. The circuit of claim **1**, wherein the threshold circuit includes a resistor having a first terminal coupled to the second terminal of the third transistor, and a second terminal coupled to the second terminal of the first transistor.

8. The circuit of claim **1**, wherein the threshold circuit includes a resistor having a first terminal coupled to the first terminal of the third transistor, and a second terminal coupled to the control terminal of the third transistor.

9. The circuit of claim **1**, wherein the threshold circuit includes a fourth transistor having a first terminal coupled to the second terminal of the third transistor, a second terminal coupled to the second terminal of the first transistor, and a control terminal coupled to the second terminal of the fourth transistor.

10. A circuit comprising:

a first transistor having a first terminal, a second terminal, and a control terminal; and

a second transistor having a first terminal coupled to the control terminal of the first transistor, a second terminal coupled to the second terminal of the first transistor, and a control terminal;

a third transistor having a first terminal coupled to the control terminal of the first transistor, a second terminal coupled to the control terminal of the second transistor, and a control terminal; and

a fourth transistor having a first terminal configured to receive a drive voltage, a second terminal coupled to the control terminal of the third transistor, and a control terminal coupled to the first terminal of the fourth transistor.

11. The circuit of claim **10**, further comprising a fifth transistor having a first terminal coupled to the control terminal of the second transistor, a second terminal coupled to the second terminal of the second transistor, and control terminal coupled to the control terminal of the third transistor.

12. The circuit of claim **10**, further comprising a resistor having a first terminal coupled to the first terminal of the second transistor and a second terminal coupled to the control terminal of the first transistor.

13. The circuit of claim **10**, further comprising a resistor having a first terminal coupled to the second terminal of the third transistor and a second terminal coupled to the control terminal of the second transistor.

14. The circuit of claim **10**, further comprising a fifth transistor having a first terminal coupled to the control terminal of the fourth transistor, a second terminal coupled to the second terminal of the second transistor, and a control terminal configured to receive a drive signal.

15. The circuit of claim **14**, further comprising:

a first resistor having a first terminal coupled to the control terminal of the fourth transistor, and a second terminal coupled to the first terminal of the fifth transistor; and

a second resistor having a first terminal coupled to the first terminal of the fourth transistor, and a second terminal coupled to the control terminal of the fourth transistor.

16. The circuit of claim **10**, further comprising a resistor having a first terminal coupled to the second terminal of the fourth transistor, and a second terminal coupled to the second terminal of the second transistor.

17. A circuit comprising:

a transistor having a first terminal, a second terminal, and a control terminal;

a driver circuit having an output coupled to the control terminal of the transistor, an input configured to receive a drive signal, and a terminal configured to receive a drive voltage; and

a transistor control circuit having a first terminal coupled to the control terminal of the transistor, a second terminal coupled to the second terminal of the transistor, and a third terminal coupled to the terminal of the driver circuit, the transistor control circuit configured to conduct current between the control terminal of the transistor and the second terminal of the transistor responsive to the drive voltage being less than a threshold.

18. The circuit of claim **17**, wherein the transistor control circuit includes a threshold circuit coupled to the third terminal of the transistor control circuit, the threshold circuit configured to:

provide a control signal having a first voltage responsive the drive voltage being less than the threshold; and

provide the control signal having a second voltage responsive to the drive voltage being greater than the threshold.

19. The circuit of claim **18**, wherein the transistor control circuit includes a drive control circuit coupled to the threshold circuit, the drive control circuit configured to conduct current between the control terminal of the transistor and the second terminal of the transistor based on the control signal having the first voltage.

20. The circuit of claim **18**, wherein:

the transistor control circuit includes a fourth terminal coupled to the input of the driver circuit; and

the threshold circuit is configured to provide the control signal having the first voltage responsive to the drive signal having a logic low state.

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